

4(a)(i)

all correct ;

1

position of nucleotide	1	2	3	4	5	6	7	8	9	10	11	12
DNA template strand	C	A	C	T	A	C	T	C	C	A	A	C
primary transcript	G	U	G	A	U	G	A	G	G	U	U	G

Question	Answer	Marks										
4(a)(ii)	all 4 correct ; <table><tr><td></td><td>aa1</td><td>aa2</td><td>aa3</td><td>aa4</td></tr><tr><td>amino acid</td><td>val / valine</td><td>met / methionine</td><td>arg / arginine</td><td>leu / leucine</td></tr></table>		aa1	aa2	aa3	aa4	amino acid	val / valine	met / methionine	arg / arginine	leu / leucine	1
	aa1	aa2	aa3	aa4								
amino acid	val / valine	met / methionine	arg / arginine	leu / leucine								
4(a)(iii)	any two from: no effect on the protein structure ; (because) all 4 triplets beginning with CA code for valine / CAA, CAG, CAT, CAC all code for valine ; idea of the genetic code is, redundant / degenerate ;	2										
4(a)(iv)	any three from: first amino acid, unchanged / still val ; changes the reading frame / described ; A.all codons (from mutation on) will change / frameshift mutation deletion alters, all amino acids after the mutation / amino acid sequence / primary structure ; ref. to, stop codon, causing premature chain termination / leading to shorter polypeptide ; (leading to) changes, in the tertiary structure / active site / AW ;	3										
4(b)	interphase <u>and</u> S phase circled ;	1										

Question	Answer	Marks
4(c)	any four from: <i>max three if points about transcription also given</i> 1 double helix unwinds, qualified ; e.g. using, enzyme / helicase breaking hydrogen bonds between strands 2 both strands act as templates ; 3 ref, activated (free DNA) nucleotides ; 4 <u>DNA polymerase</u>, plus example of role ; e.g. adds complementary nucleotides to exposed strand forms phosphodiester bonds (between adjacent nucleotides) proofreading / checking for errors / checking for mismatches 5 leading strand synthesised continuously / AW ; 6 lagging strand synthesised in (Okazaki) fragments ; 7 ligase connects (lagging strand), fragments / nucleotides, (with phosphodiester bonds) ; 8 semi-conservative replication / AW ; e.g. both new double helices have one, parental / conserved, strand and one newly synthesised strand	4
4(d)	ribose ; I pentose	1

Question	Answer	Marks
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